

⑦

MATH TEST

1) $P(x) = 3x^2 + 36x + 5$

$$\frac{dP}{dx} = 6x + 36$$

$$6(5) + 36$$

$$(c) 66$$

2) $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$

$$\sin^{-1}\left(\sin\left(\pi - \frac{\pi}{3}\right)\right)$$

$$\sin^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$$

$$(b) \frac{\pi}{3}$$

3) $|A| = 4$

$$|kA| = k^n |A|$$

$$2^3 |A| = 8 \times 4 = 32$$

$$(c) 32$$

4) $P(A) = 0.8$ $P(B/A) = 0.4$

$$P(B) = 0.5$$

$$P(A \cap B) = P(A) \cdot P(B/A)$$

$$= 0.8 \times 0.4$$

$$(a) 0.32$$

5) $u = \hat{i} + \hat{j} + \hat{k}$ & $v = \hat{i} - \hat{j} + \hat{k}$

$$u \times v =$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}$$

$$= \hat{i}(1+1) - \hat{j}(1-1) + \hat{k}(-1-1)$$

$$= 2\hat{i} - 2\hat{k}$$

$$\text{Magnitude} = \sqrt{2^2 + 0^2 + (-2)^2}$$

$$= \sqrt{8}$$

$$= 2\sqrt{2}$$

6) (b) The region extends infinitely in atleast 1 direction

7) (b) 8

8) $\log(\sec x + \tan x)$

(a) $\sec x$

9) (d) Equivalence

10) (b) $B^T A^T$

11) $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{2\sqrt{3}}{4\sqrt{3}} = \frac{1}{2}$

(b) $\theta = \frac{\pi}{3}$

12)

$$3x - 4 = y$$

$$2x = y + 4$$

$$x = \frac{y+4}{2}$$

(a) $\frac{x+4}{3}$

13)

$$\frac{4}{52} \times \frac{3}{51} = \frac{1}{85}$$

(b) $\frac{1}{221}$

14)

(d) $-\tan^{-1} 1$

15)

(b) non-negativity constraints

16)

(a)

17)

(a)

18) (b)

19) (d)

20) (c)

21) $\vec{p} = 2\hat{i} - \hat{j} + 3\hat{k}$ $\vec{q} = \hat{i} + \hat{j} + \hat{k}$

(i) $\vec{q} - \vec{p} = \hat{i} + \hat{j} + \hat{k} - (2\hat{i} - \hat{j} + 3\hat{k}) = -\hat{i} + 2\hat{j} - 2\hat{k}$

Distance $|\vec{pq}| = \sqrt{(-1)^2 + 2^2 + (-2)^2}$
 $= \sqrt{1+4+4}$
 $= 3 \text{ units}$

(ii) $\vec{OP} = 2\hat{i} - \hat{j} + 3\hat{k}$ $\vec{OQ} = \hat{i} + \hat{j} + \hat{k}$

$|\vec{OP}| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4+1+9} = \sqrt{14}$

$|\vec{OQ}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$\vec{OP} \cdot \vec{OQ} = (2)(1) + (-1)(1) + (3)(1)$
 $= 4$

$\cos \theta = \frac{\vec{OP} \cdot \vec{OQ}}{|\vec{OP}| |\vec{OQ}|} = \frac{4}{\sqrt{14} \times \sqrt{3}} = \frac{4}{\sqrt{42}}$

22) $M_1: x + y + 2z = 10$

$M_2: x + 2y + 3z = 16$

$M_3: x + 3y + z = 14$

(i) $AX = B$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 16 \\ 14 \end{bmatrix}$$

$X = A^{-1}B$

(ii) $|A| = 1(2-9) - 1(1-3) + 2(3-2) = -7 + 2 + 2 = -3$

4

$$\text{adj } A = \begin{bmatrix} -7 & 2 & 1 \\ 5 & -1 & -2 \\ -1 & -1 & 1 \end{bmatrix}^T = \begin{bmatrix} -7 & 5 & -1 \\ 2 & -1 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{-1}{3} \begin{bmatrix} -7 & 5 & -1 \\ 2 & -1 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} = \frac{-1}{3} \begin{bmatrix} -7 & 5 & -1 \\ 2 & -1 & -1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 16 \\ 14 \end{bmatrix}$$

$$= -\frac{1}{3} \begin{bmatrix} -70 + 80 - 14 \\ 20 - 16 - 14 \\ 10 - 32 + 14 \end{bmatrix}$$

$$= -\frac{1}{3} \begin{bmatrix} -4 \\ -10 \\ -8 \end{bmatrix}$$

$$\boxed{x = \frac{4}{3}}, \boxed{y = \frac{10}{3}}, \boxed{z = \frac{8}{3}}$$

∴ Unique soln exists

22) $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ $A^T = \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix}$

Symmetric,

$$P = \frac{1}{2} (A + A^T) = \frac{1}{2} \begin{bmatrix} 6 & -3 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -3/2 \\ -3/2 & -1 \end{bmatrix}$$

Skew-symmetric,

$$Q = \frac{1}{2} (A - A^T) = \frac{1}{2} \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -5/2 \\ 5/2 & 0 \end{bmatrix}$$

$$A = P + Q$$

$$= \begin{bmatrix} 3 & -3/2 \\ -3/2 & -1 \end{bmatrix} + \begin{bmatrix} 0 & -5/2 \\ 5/2 & 0 \end{bmatrix}$$

5

$$24) \quad 2 \tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right) = \frac{\pi}{4}$$

$$\text{LHS} \quad \tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right)$$

$$\tan^{-1} \left(\frac{2/3}{8/9} \right) + \tan^{-1} \left(\frac{1}{7} \right)$$

$$\tan^{-1} \left(\frac{6}{8} \right) + \tan^{-1} \left(\frac{1}{7} \right)$$

$$= \tan^{-1} \left(\frac{3}{4} \right) + \tan^{-1} \left(\frac{1}{7} \right)$$

$$= \tan^{-1} \left(\frac{25/28}{25/28} \right) = \tan^{-1} (1)$$

$$= \frac{\pi}{4} //$$

$$25) \quad f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \pi/2 \\ 3, & \text{if } x = \pi/2 \end{cases} \quad \text{at } x = \frac{\pi}{2}$$

$$\lim_{x \rightarrow \pi/2} \frac{k \cos x}{\pi - 2x} = f(\pi/2) = 3$$

$$\text{let } x = \frac{\pi}{2} + h. \text{ As } x \rightarrow \pi/2, h \rightarrow 0$$

$$\lim_{h \rightarrow 0} \frac{k \cos(\pi/2 + h)}{\pi - 2(\pi/2 + h)} = \lim_{h \rightarrow 0} \frac{-k \sin h}{-2h} = \frac{k}{2}$$

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = \frac{k}{2}$$

$$f\left(\frac{\pi}{2}\right) = \frac{k}{2} = 3$$

$$\therefore \boxed{k = 6}$$

28) Let x kg of food 2 & y kg of food 1 be mixed.

Minimize, $Z = 50x + 70y$

Constraints:

Vitamin A: $2x + y \geq 8$

Vitamin C: $x + 2y \geq 10$

Non-negativity: $x \geq 0, y \geq 0$

29) 5R & 5B $\Rightarrow$ (Total = 10)

Case I

First ball is Red, $P(R) = \frac{5}{10} = \frac{1}{2}$

Add 2 Red $\rightarrow$ urn has 7 Red, 5 Black (Total = 12)

$P(R_2/R_1) = \frac{7}{12}$

Case II

First ball is black $\left(P(B) = \frac{5}{10} = \frac{1}{2} \right)$

Add 2 Black $\rightarrow$ urn has 5 Red, 7 Black (Total = 12)

$P(B_2/B_1) = \frac{5}{12}$

$$P(R_2) = P(R_1)P(R_2/R_1) + P(B_1)P(R_2/B_1)$$

$$= \frac{1}{2} \times \frac{7}{12} + \frac{1}{2} \times \frac{5}{12} = \frac{12}{24} = \frac{1}{2}$$

30)

Reflexivity

Transitivity: For all $a \in A$, $|a-a| = 0$, which is even

So, $(a,a) \in R$ $\therefore$ Reflexive

Symmetric: If $(a,b) \in R$, $|a-b|$ is even $\Rightarrow |-(b-a)| = |b-a|$ is even

So, $(b,a) \in R$ $\therefore$ Symmetric

Transitivity: If $(a,b) \in R$ & $(b,c) \in R$, then $a-b = 2k_1$ and $b-c = 2k_2$, $a-c = 2(k_1+k_2) \Rightarrow (a,c) \in R$

$\therefore$ Transitive

$\therefore$ Equivalence relation.

31)

$$x - y + z = 4$$

$$2x + y - 3z = 0$$

$$x + y + z = 2$$

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1}B$$

$$|A| = 1(1+3) + 1(2+3) + 1(2-1)$$

$$= 4 + 5 + 1 = 10$$

$$\text{adj } A = \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}^T = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$X = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 16 + 0 + 4 \\ -20 + 0 + 10 \\ 4 + 0 + 6 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 20 \\ -10 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\therefore \boxed{x=2}, \boxed{y=-1}, \boxed{z=1}$$

8

32) $V = \frac{4}{3} \pi r^3$

34) Total drivers = 2000 + 4000 + 6000
= 12000

Let E_1, E_2, E_3 be events that he is a scooter, car, truck driver respectively.

$$P(E_1) = \frac{2000}{12000} = \frac{1}{6}$$

$$P(E_2) = \frac{4000}{12000} = \frac{1}{3}$$

$$P(E_3) = \frac{6000}{12000} = \frac{1}{2}$$

Let A be event that driver meets with accident

$$P(A/E_1) = 0.01 = \frac{1}{100}$$

$$P(A/E_2) = 0.03 = \frac{3}{100}$$

$$P(A/E_3) = 0.15 = \frac{15}{100}$$

$$P(E_1/A) = \frac{P(E_1) P(A/E_1)}{P(E_1) P(A/E_1) + P(E_2) P(A/E_2) + P(E_3) P(A/E_3)}$$

$$= \frac{\frac{1}{6} \times \frac{1}{100}}{\frac{1}{6} \times \frac{1}{100} + \frac{1}{3} \times \frac{3}{100} + \frac{1}{2} \times \frac{15}{100}}$$

$$= \frac{\frac{1}{600}}{(1 + 6 + 45)/600}$$

$$= \frac{1}{52}$$

35) $f: \mathbb{R}^+ \rightarrow [4, \infty)$ $f(x) = x^2 + 4$ $\left(\frac{2}{\sqrt{17}}\right) \in$

One-one

Let $x_1, x_2 \in \mathbb{R}^+$ such that $f(x_1) = f(x_2)$

$$\Rightarrow x_1^2 + 4 = x_2^2 + 4$$

$$x_1^2 = x_2^2$$

$$\boxed{x_1 = x_2}$$

Hence, f is one-one.

Onto

Let $y \in [4, \infty)$

$$y = x^2 + 4$$

$$x^2 = y - 4$$

$$x = \sqrt{y-4}$$

Since $y \geq 4$, $y-4 \geq 0$,
every element has a pre-image

$\therefore f$ is onto

$\therefore f$ is bijective, thus, invertible.

$$f^{-1}(x) = \sqrt{x-4} //$$

36)

$$Z = 5x + 3y$$

Subject to constraints:

$$8x + 5y \leq 15$$

$$5x + 2y \leq 10$$

$$x \geq 0, y \geq 0$$

$O(0,0)$

$$\begin{array}{r} 6x + 10y \leq 30 \\ \Leftrightarrow 25x + 10y \leq 50 \\ \hline -19x \geq -20 \end{array}$$

$$\boxed{x \leq \frac{20}{19}}$$

$$3\left(\frac{20}{19}\right) + 5y = 15$$

$$5y = 15 - \frac{60}{19}$$

$$5y = \frac{225}{19}$$

$$y = \frac{45}{19}$$

$$A = (2, 0)$$

$$B = \left(\frac{20}{19}, \frac{45}{19}\right)$$

$$C = (0, 3)$$

~~A, B, C~~

$$\text{At } O(0, 0), Z = 0$$

$$\text{At } A(2, 0), Z = 10$$

$$\text{At } B; Z = 12.37$$

$$\text{At } C, Z = 9$$

∴ Max value for
 $Z = 12.37$ at $\left(\frac{20}{19}, \frac{45}{19}\right)$

37) $f(x) = x^2 - 4x + 3$ on $[1, 3]$

Condition 1: $f(x)$ is a polynomial fnt.

∴ continuous on closed interval $[1, 3]$.

Condition 2: $f'(x) = 2x - 4$ exists for all $x \in [1, 3]$